



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 06

DHCP, DNS, Email, and Web Server Configuration

■ Objectives

- (a) Learn how to configure a DHCP server.
- (b) Learn how to configure a DNS server.
- (c) Learn how to configure a Web server.
- (d) Learn how to configure an Email server.

■ Requirements

- (a) Cisco Packet Tracer software.
- (b) Two routers and three switches.
- (c) Four PCs and one network sniffer.
- (d) Four servers.

■ Pre-Lab Task

Students are required to build the network topology shown in Figure 6, following the guidelines provided in Section 6.6.1. In addition, they must configure the IP addresses and OSPF routing protocol as described in Section 6.6.2 (A)–(C).

6.1 Introduction

In modern networks, several core services work together to enable seamless communication and access to resources. One of the key components is the *Dynamic Host Configuration Protocol (DHCP)*, which automates the process of assigning *IP addresses* and network configuration settings to devices. This allows hosts to connect to the network and use essential services without manual setup. *DHCP* also enables access to services that rely on *IP-based communication*, including the *Domain Name System (DNS)*, the *Network Time Protocol (NTP)*, and applications based on the *User Datagram Protocol (UDP)* or *Transmission Control Protocol (TCP)*.

The *DNS* acts as the Internet's phonebook by translating human-friendly domain names (such as *birzeit.edu* or *nytimes.com*) into machine-readable *IP addresses*, enabling web browsers and other services to locate and connect to servers efficiently.

A *web server* is a combination of software and hardware designed to deliver web content over the Internet using the *Hypertext Transfer Protocol (HTTP)* or its secure version, *HTTPS*. Similarly, an *email server* is responsible for sending, receiving, and storing email messages. It typically uses the *Simple Mail Transfer Protocol (SMTP)* for sending emails and the *Post Office Protocol (POP3)* or the *Internet Message Access Protocol (IMAP)* for retrieving them. Secure communication is often ensured through encryption protocols such as *SSL/TLS*.

In this experiment, we will configure and simulate *DHCP*, *DNS*, *web*, and *email servers* using *Cisco Packet Tracer*. The following sections will explain each service and demonstrate how to set them up effectively within a *virtual network environment*.

6.2 Dynamic Host Configuration Protocol (DHCP)

DHCP is a system for assigning *IP addresses* to each network device (known as a *host*) on an organization's network. A *host* may be a desktop computer, a laptop, a tablet, a mobile device, a thin client, or other types of device. Each host must have an *IP address* to communicate with other devices over the Internet. The *DHCP* network protocol assigns addresses automatically, rather than requiring network administrators to make manual assignments. *DHCP* is also responsible for automatically assigning new *IP addresses* when devices move to new locations on the network. In addition to *IP addresses*, the *DHCP* service assigns configuration parameters such as *DNS addresses*, *subnet masks*, and *default gateways* that are essential to network communications.

6.2.1 How does DHCP work?

DHCP operates at the application layer and uses the UDP transport protocol. Its primary function is to dynamically assign IP addresses and provide essential TCP/IP configuration information to clients. DHCP is a client-server protocol; it uses port 67 for the server and port 68 for the client. When a new host joins the network, the DHCP process follows a four-step sequence, as shown in Figure 1:

1. **DHCP Discover:** The DHCP client broadcasts a DHCP Discover message to locate available DHCP servers. This message is sent with a destination IP address of 255.255.255.255 (broadcast) and a source IP address of 0.0.0.0 (since the

client has no IP address yet). The message is encapsulated in a datagram and sent over the link layer to reach all nodes on the subnet.

2. **DHCP Offer:** A DHCP server that receives the discover message responds with a DHCP Offer message. This message is also broadcast using 255.255.255.255 and contains the transaction ID from the discover message, a proposed IP address for the client, subnet mask, lease duration (the amount of time for which the IP address will be valid), and other configuration parameters.
3. **DHCP Request:** The client responds to the selected offer with a DHCP Request message, indicating acceptance of the proposed configuration. This message is also broadcast to inform other DHCP servers that their offers were declined.
4. **DHCP Acknowledgment (ACK):** The selected DHCP server replies with a DHCP ACK message, confirming the IP address assignment and finalizing the configuration. Once this message is received, the client can begin using the assigned IP address for the duration of the lease. If needed, the client can later request a lease renewal to extend the IP address usage.

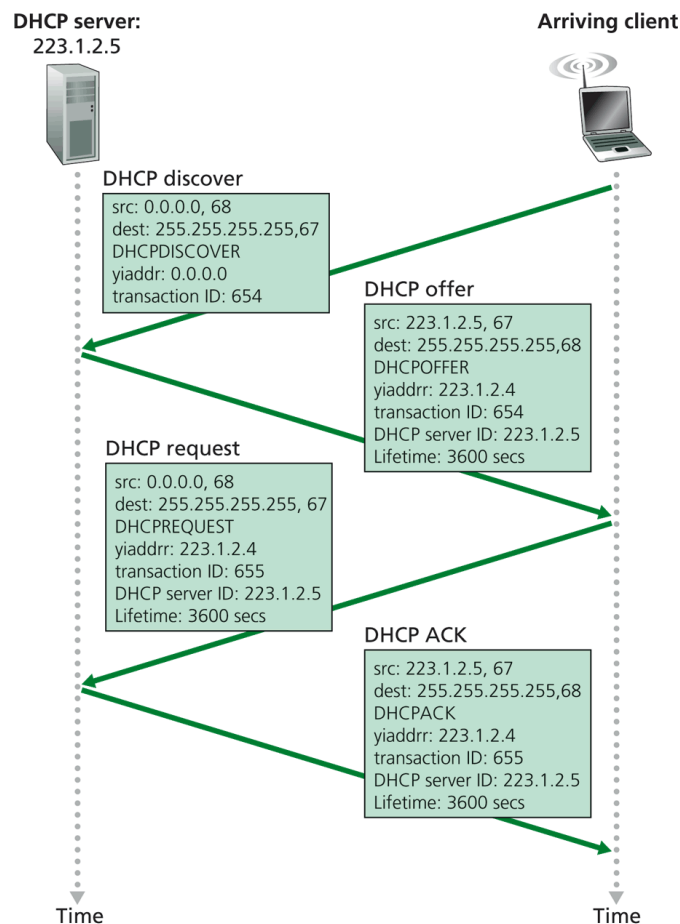


Figure 1: DHCP client-server interaction.

6.2.2 Benefits of Using DHCP

- DHCP automates and simplifies the assignment of IP addresses, reducing the administrative burden on network managers.

- By dynamically allocating and reusing IP addresses, DHCP helps optimize address usage, reducing the total number of addresses required.
- Organizations can modify IP address schemes seamlessly without disrupting end users, making network transitions smoother.
- DHCP minimizes human errors by centralizing and automating IP address assignments, preventing duplicate addresses and misconfigurations.

6.2.3 DHCP Configuration on a Router

To configure a Cisco router to provide DHCP services for a subnet, follow these steps:

1. **Exclude Reserved IP Addresses:** Before creating a DHCP pool, reserve specific IP addresses that should not be assigned automatically by DHCP. These addresses are typically used for router interfaces (default gateway), servers (e.g., DNS, Web, File servers), and network devices requiring static IPs. To exclude an IP range from being assigned dynamically, use:

```
Router(config)# ip dhcp excluded-address <FIRST-ADDRESS>  
                        <LAST-ADDRESS>
```

2. **Create a DHCP Pool:** Next, define a DHCP pool for the subnet. This requires specifying the network range, default gateway, and DNS server:

```
Router(config)# ip dhcp pool <POOL-NAME>  
Router(dhcp-config)# network <NETWORK-ID> <SUBNET-MASK>  
Router(dhcp-config)# default-router <GATEWAY-IP-ADDRESS>  
Router(dhcp-config)# dns-server <DNS-SERVER-IP-ADDRESS>
```

3. **Activate DHCP Service:** Finally, make sure the DHCP service is enabled on the router:

```
Router(config)# service dhcp
```

6.3 Domain Name System (DNS)

DNS translates human-readable *domain names* into *IP addresses*, which browsers use to load web pages. Every device connected to the Internet has a unique *IP address*, which other devices use to communicate with it. *DNS* eliminates the need for users to memorize complex numerical *IP addresses* by allowing them to enter easy-to-remember *domain names* instead.

6.3.1 DNS Server

A *DNS server* is a computer system that maintains a database of public *IP addresses* and their associated *domain names*. When a user enters a *domain name*—such as google.com—into a web browser, the *DNS server* resolves that name to its corresponding *IP address* (i.e., 142.250.72.206). This *IP address* directs the user's device to the correct server hosting the website. *DNS* functions much like a phonebook for the Internet, enabling efficient and user-friendly access to websites.

6.3.2 Distributed, Hierarchical DNS

The *DNS infrastructure* is distributed and hierarchical, consisting of many servers world-wide. No single server holds all *domain name-to-IP address* mappings. Instead, the information is divided and managed across different types of *DNS servers*, as shown in Figure 2. The three main classes of *DNS servers* are:

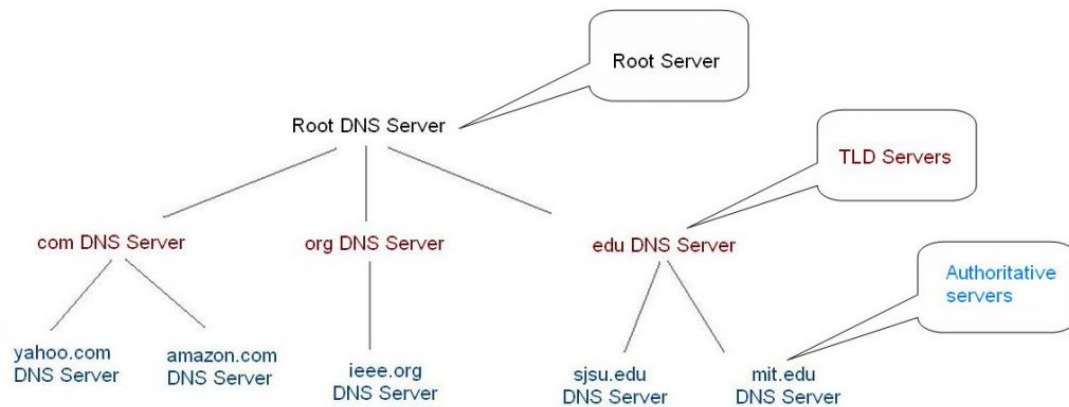


Figure 2: DNS server hierarchy.

- **Root DNS Servers:** These are the highest level in the DNS hierarchy. The root server system consists of 1916 instances operated by 12 independent organizations, illustrated in Figure 3. Root servers respond to queries by providing the IP addresses of Top-Level Domain (TLD) servers.

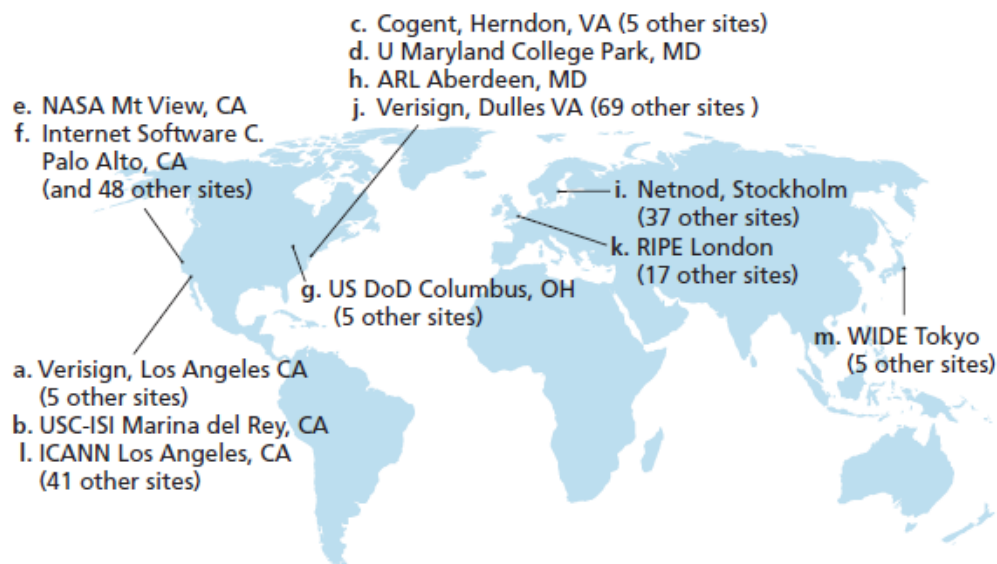


Figure 3: DNS root servers worldwide.

- **Top-Level Domain (TLD) Servers:** Each top-level domain—such as *.com*, *.org*, *.net*, *.edu*, or country-specific domains like *.uk*, *.fr*, *.ca*, and *.jp*—has a dedicated

TLD server (or server cluster). TLD servers respond with the IP addresses of authoritative DNS servers responsible for specific domain names.

- **Authoritative DNS Servers:** These servers store and serve the actual DNS records for domain names. Any organization that operates publicly accessible services, such as web or email servers, must maintain authoritative DNS records mapping hostnames to IP addresses. Most large companies and universities operate their own primary and secondary (backup) authoritative DNS servers to ensure reliability and control.

6.3.3 Overview of How DNS Works

DNS primarily uses *UDP Port 53*. However, as the internet evolves and security and functionality demands grow, DNS is increasingly relying on *TCP Port 53*. From its inception, DNS was designed to use both UDP and TCP: UDP is the default due to its lower overhead, while TCP is used as a fallback when a response is too large to fit in a single UDP packet or when reliability is required.

To understand how DNS resolution works in practice, consider a scenario in which a host at `engineering.nyu.edu` needs to resolve the domain name `gaia.cs.umass.edu` into an IP address. This process can be carried out using either *iterative* or *recursive* queries, as illustrated in Figure 4. In **recursive queries**, the DNS server takes full responsibility for resolving the domain on behalf of the client, performing all necessary lookups and returning either a final answer or an error. In **iterative queries**, the DNS server returns the best information it has—typically a referral to another server—and leaves it to the client or resolver to continue the process.

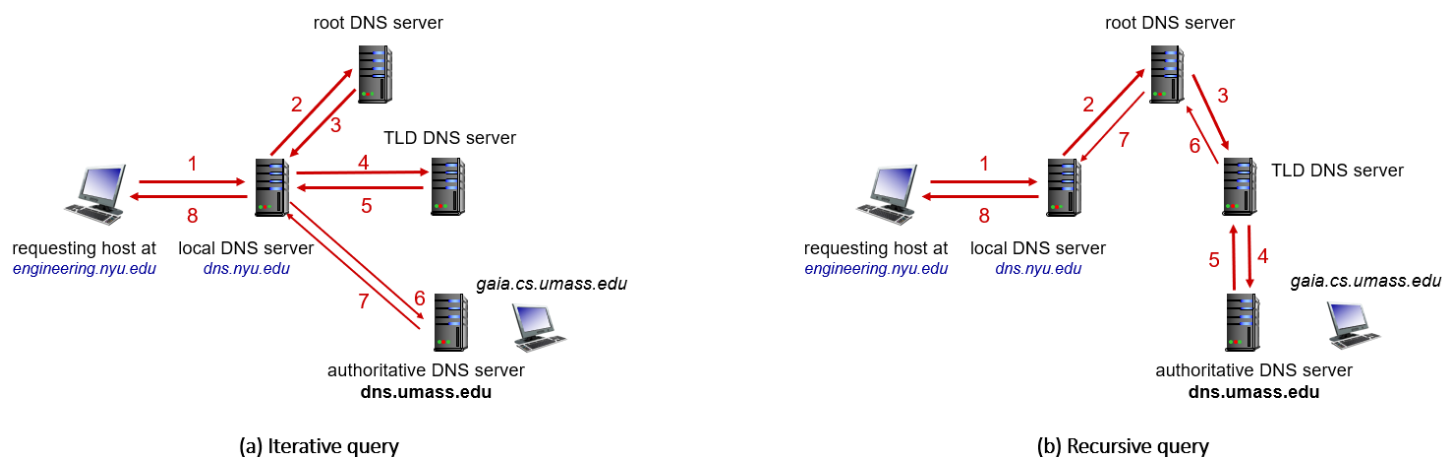


Figure 4: Types of DNS query resolution.

In practice, a local DNS resolver typically performs recursive queries on behalf of end users, while itself using iterative queries to walk through the DNS hierarchy. This hybrid resolution process is shown in Figure 4(a) and proceeds as follows:

1. The user's computer (in this case, the host at `engineering.nyu.edu`) sends a DNS query to the local DNS server (`dns.nyu.edu`), often provided by the ISP.
2. If the local DNS server has the requested IP address cached, it replies directly to the user. If not, it initiates a query to a root DNS server.

3. The root server responds with a referral to a TLD DNS server responsible for `.edu` domains.
4. The local DNS server then queries the TLD server.
5. The TLD server responds with a referral to the authoritative DNS server for `cs.umass.edu`.
6. The local DNS server queries the authoritative server (`dns.umass.edu`).
7. The authoritative server returns the IP address for `gaia.cs.umass.edu`.
8. The local DNS server forwards the resolved IP address to the requesting host.

6.4 Web Server

A *web server* refers to both the software and hardware that handle client requests over the *World Wide Web* using the *HTTP*, which operates on *TCP port 80*, and its secure version, *HTTPS*, which operates on *TCP port 443*. The primary function of a web server is to store, process, and deliver *web pages* to users, enabling the display of website content in a browser. In addition to HTTP, web servers may also support other protocols such as *SMTP*, used for email communication over *TCP port 25*, and *File Transfer Protocol* (FTP), used for file transfers over *TCP port 21*. Web servers play a critical role in web hosting, providing the infrastructure needed to host websites and *web-based applications*.

6.4.1 Web Client-Server Interaction

- When a user enters a web address such as `www.example.com` into a browser and presses Enter, the browser begins the process of retrieving the web page.
- First, the domain name is resolved to an IP address using the DNS.
- Once the IP address is obtained, the browser initiates a **TCP connection** with the web server using the **TCP three-way handshake**:
 - The client sends a **SYN** (synchronize) packet to the server.
 - The server responds with a **SYN-ACK** (synchronize-acknowledge) packet.
 - The client sends an **ACK** (acknowledge) packet.
- After the reliable connection is established, the browser constructs an **HTTP request** (such as a **GET** request) and sends it to the server over the TCP connection.
- The client's ISP routes this request through the Internet to the server hosting `www.example.com`.
- The server receives the request, processes it, and generates an **HTTP response** (e.g., containing an HTML page).
- The response is sent back through the server's ISP, over the Internet, and eventually reaches the client's machine.
- The browser receives the HTTP response, parses the content (such as HTML, CSS, and JavaScript), and renders the web page for the user to view.

6.5 Email Server

An email server is a system responsible for the sending, receiving, storing, and management of email messages. It ensures the reliable delivery of messages between senders and recipients and operates continuously to allow email communication at any time. Email servers are widely used by individuals, businesses, and organizations to facilitate effective communication over the internet or within private networks. Some organizations choose to deploy their own email servers to maintain better security and control, while others opt for cloud-based services provided by companies such as Google or Microsoft.

To send emails, servers use the *SMTP*. When a user sends an email, the originating server processes the message and forwards it to the recipient's email server using SMTP. Upon arrival, the recipient's server stores the email until the user retrieves it. For email retrieval, servers typically employ either *POP3* or *IMAP*. POP3 downloads the email to the user's device and subsequently removes it from the server, making it suitable for devices with limited storage capacity. In contrast, IMAP retains messages on the server, allowing users to access and manage their emails from multiple devices while maintaining synchronization. Figure 5 illustrates the general flow of messages between email servers.

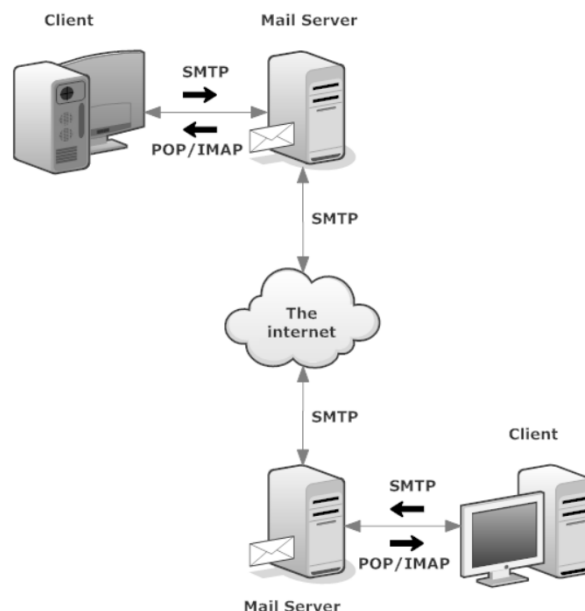


Figure 5: Mail server message flow.

To ensure that emails are correctly routed to their destinations, email servers rely on the *DNS* and *Mail Exchange* (MX) records. These records determine the appropriate destination server based on the domain name in the recipient's email address (e.g., @gmail.com). In addition to routing, email servers incorporate various authentication and security mechanisms to protect against spam, phishing, and email spoofing. Modern email services implement encryption to ensure secure transmission of messages. The standard (non-secure) port numbers are as follows: IMAP uses port 143, POP3 uses port 110, and SMTP uses port 25. For secure communication, IMAP over SSL/TLS operates on port 993, POP3 over SSL/TLS uses port 995, and SMTP over SSL/TLS typically uses port 465 or port 587 when employing STARTTLS.

6.6 Procedure

In this lab, we will connect several PCs, routers, and multiple servers on different networks. This will require configuring key network services: DHCP, DNS, Email, and a Web server. These services will provide dynamic IP addressing, domain name resolution, and web service accessibility to devices on the network. The email server will handle sending and receiving emails within the network. Additionally, we will configure *Open Shortest Path First* (OSPF) to enable routing between routers and allow communication between all subnets.

6.6.1 Building the Topology

Construct the network topology shown in Figure 6, which consists of the following devices:

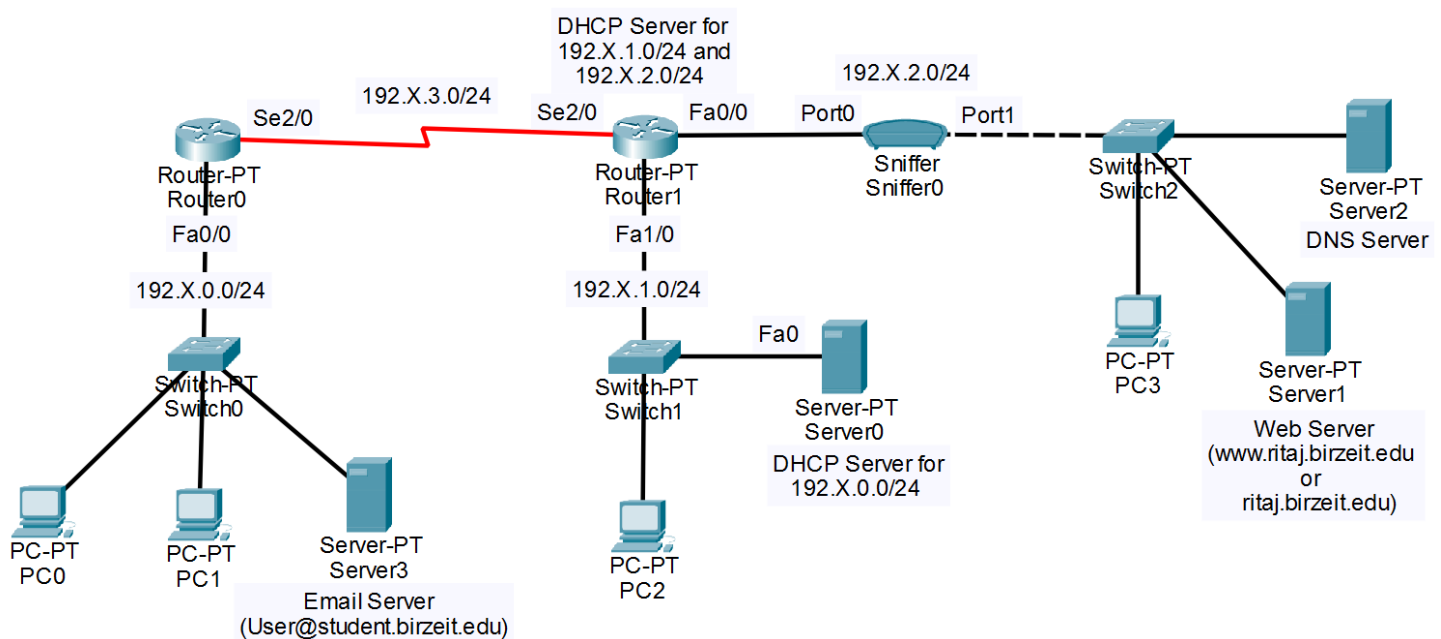


Figure 6: Server configuration topology.

- Routers: Use **Router-PT**.
 - Router0: This router will run OSPF routing in Area 0.
 - Router1: This router will function as the DHCP server for the two subnets (192.X.1.0/24 and 192.X.2.0/24) and will also run OSPF routing in Area 0.
- Switches: Use **Switch-PT**.
- Servers: Use **Server-PT**.
- PCs: Use **PC-PT**.
 - PC0 and PC1 will receive dynamic IP addresses from the DHCP Server (Server0).
 - PC2 and PC3 will receive dynamic IP addresses from Router1 that act as a DHCP Server.
- Packet Capture: Place a **Sniffer** between Router1 and Switch2 to effectively monitor incoming packets.

- Connections: Use the “**Automatically Choose Connection Type**” option to link PCs, switches, routers, servers, and sniffer.

6.6.2 Network Setup and Configuration

A) Configuring the Static IPs for Routers Interfaces

First, configure static IP addresses on the routers' interfaces. The value of **X** in the network address corresponds to the *last two digits* of your student ID. For example, if the network address is **192.X.0.0/24** and your student ID is **1230105**, then **X = 05**, resulting in the network address **192.5.0.0/24**.

- For **Router0**, configure static IP addresses on:

- Fa0/0 interface (connected to Switch0, 192.X.0.0/24 network):

```
Router(config)# interface fa0/0
Router(config-if)# ip address 192.X.0.1 255.255.255.0
Router(config-if)# no shutdown
```

- Se2/0 interface (connected to Router1, 192.X.3.0/24 network):

```
Router(config)# interface se2/0
Router(config-if)# ip address 192.X.3.1 255.255.255.0
Router(config-if)# no shutdown
```

- For **Router1**, configure static IP addresses on the relevant interfaces using the appropriate network addresses, following the same steps as Router0.

B) IP Configuration for the Servers

Figures 7 to 10 illustrate the IP configurations for the DNS, DHCP, web, and email servers. These configurations include the static IP addresses, subnet masks, default gateways, and DNS settings assigned to each server.

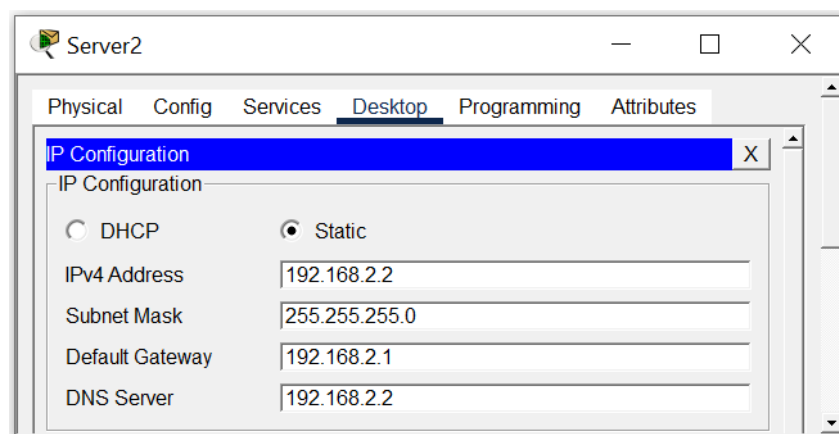


Figure 7: IP configuration for DNS server.

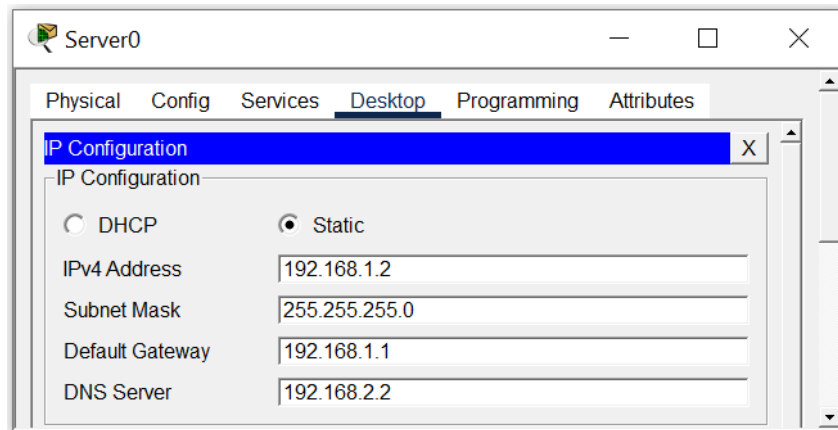


Figure 8: IP configuration for DHCP server.

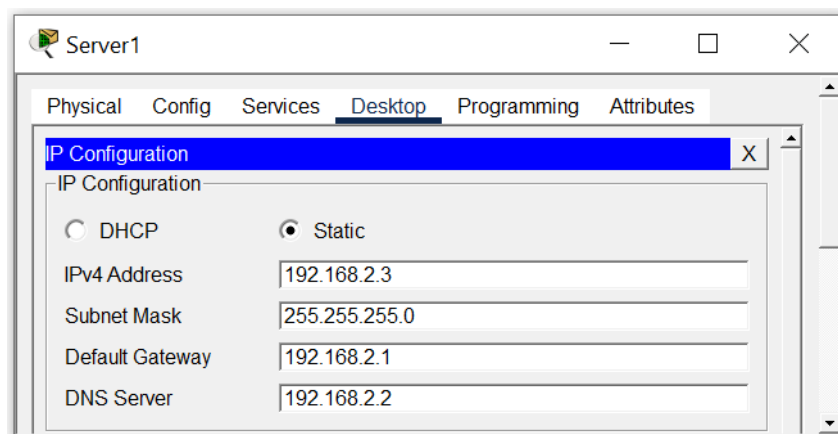


Figure 9: IP configuration for web server.

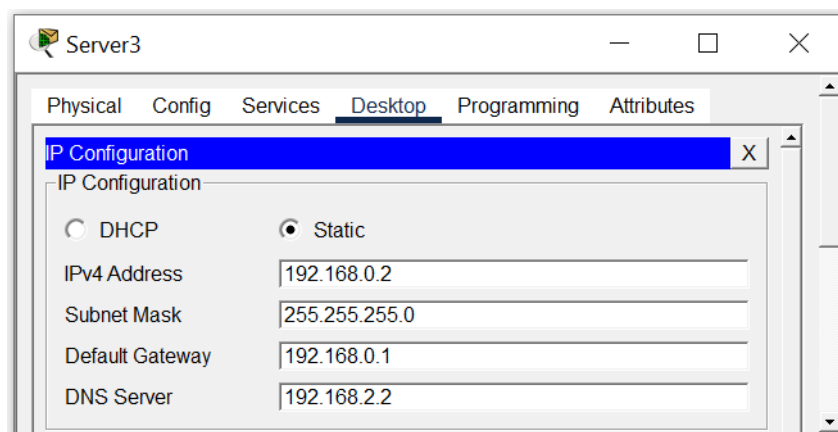


Figure 10: IP configuration for email server.

C) Enabling the OSPF Routing

Enable OSPF routing on both **Router0** and **Router1**, as described in previous experiments. For example, to configure OSPF on **Router0**, use the following commands:

```
Router(config)# router ospf 1
Router(config-router)# network 192.X.0.0 0.0.0.255 area 0
Router(config-router)# network 192.X.3.0 0.0.0.255 area 0
```

Now, apply a similar configuration on **Router1**, adjusting the network statements according to its connected networks.

D) Configuring the DHCP Service on Router1

We will configure **Router1** to serve as the DHCP server for the 192.X.1.0/24 and 192.X.2.0/24 subnets. Configuration for network **192.X.1.0/24**:

1. **Exclude Reserved IP Addresses:** Exclude the first 10 IP addresses (192.X.1.1 to 192.X.1.10) to reserve them for the default gateway, Server0, and future use:

```
Router(config)# ip dhcp excluded-address 192.X.1.1
                  192.X.1.10
```

2. **Create DHCP Pool:** Set up the DHCP pool for the subnet, specifying the network, default gateway, and DNS server:

```
Router(config)# ip dhcp pool LAN1
Router(dhcp-config)# network 192.X.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.X.1.1
Router(dhcp-config)# dns-server 192.X.2.2
Router(dhcp-config)# exit
```

3. **Activate DHCP Service:** Ensure the DHCP service is enabled on the router:

```
Router(config)# service dhcp
```

Now, apply a similar configuration for the **192.X.2.0/24** subnet by creating another DHCP pool and excluding the reserved IP addresses for that network.

E) Configuring the DHCP Server

We will now configure **Server0** to dynamically assign IP addresses to the 192.X.0.0/24 network, as illustrated in Figure 11. The configuration will be performed using the following steps:

1. Open **Server0** and navigate to the **Services** tab.
2. Select **DHCP** from the left-hand menu.
3. Select the appropriate interface (**FastEthernet0** in this case).
4. Ensure the Service is turned **On**.
5. In the **Pool Name** field, enter a name (e.g., Pool1).
6. Set the **Default Gateway** to 192.X.0.1, which will act as the gateway for the assigned addresses.

7. Enter the **DNS Server** address as 192.X.2.2, which will handle domain name resolution.
8. Set the **Start IP Address** to 192.X.0.3, since 192.X.0.1 and 192.X.0.2 are reserved for the default gateway and Server3, respectively.
9. Click **Add** to add the DHCP pool to the server's settings.
10. Click **Save** to finalize the configuration.

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Pool1

Default Gateway: 192.168.0.1

DNS Server: 192.168.2.2

Start IP Address: 192.168.0.3

Subnet Mask: 255.255.255.0

Maximum Number of Users: 253

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Pool1	192.168.0.1	192.168.2.2	192.168.0.3	255.255.255.0	253	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.0	255.255.255.0	255	0.0.0.0	0.0.0.0

☐ Top

Figure 11: Configuring the DHCP server.

Once configured, DHCP clients (PCs) in network 192.X.0.0/24 will receive an IP address dynamically from the defined pool, along with the appropriate gateway and DNS settings.

F) Configuring DHCP Relay on Router0

Since the DHCP server (**Server0**) is not directly connected to the 192.X.0.0/24 network, we must configure **Router0** to forward DHCP broadcast messages from this subnet to **Server0** (IP address: 192.X.1.2). This is achieved using the **ip helper-address** command. On **Router0**, navigate to the FastEthernet0/0 interface (connected to the 192.X.0.0/24 network) and enter the following commands:

```
Router(config)# interface fa0/0
Router(config-if)# ip helper-address 192.X.1.2
```

G) Assigning Dynamic IP Addresses

To assign dynamic IP addresses to *all PCs*, follow these steps:

- On each PC, go to **Desktop**.
- Set the IP Configuration to **DHCP**.

On **PC0** and **PC1**, verify that they receive the correct IP configuration — including the IP address, subnet mask, default gateway, and DNS server — from the DHCP server. Figure 12 shows the DHCP IP configuration for **PC0**. Similarly, on **PC2** and **PC3**, verify that they obtain the correct IP configuration from the router's DHCP service. Figure 13 shows the DHCP IP configuration for **PC3**.

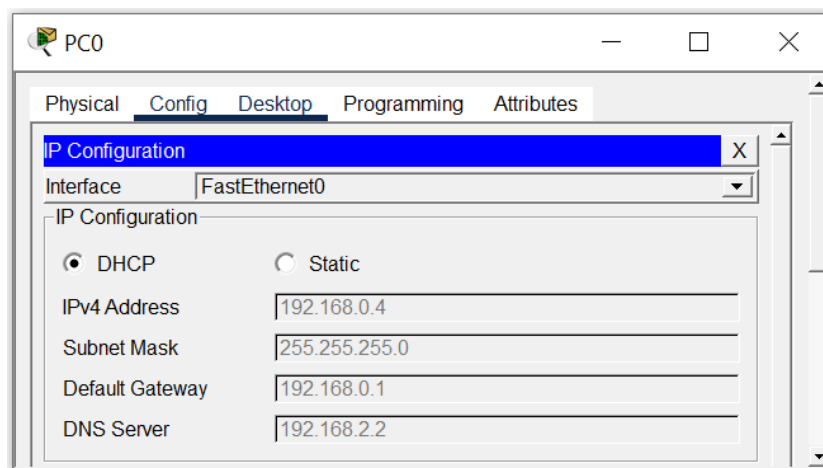


Figure 12: IP configuration for PC0.

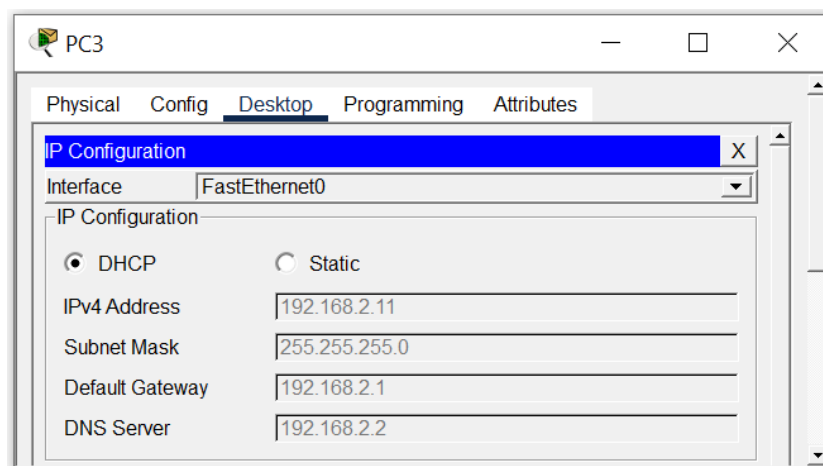


Figure 13: IP configuration for PC3.

H) Configuring the Web Server

In this step, you will configure **Server1** to act as a web server and host a sample webpage by following the steps below:

1. Open **Server1** and navigate to the **Services** tab.

2. Under Services, select **HTTP** and switch the HTTP service to **On**.
3. In the **File Manager**, update the default webpage by modifying the *index.html* file and importing all necessary files. For this lab, the goal is to have the server host the **Ritaj webpage** (Birzeit University's Academic and Administrative Portal).
 - Open ritaj.birzeit.edu in your web browser.
 - Right-click anywhere on the page and select “**Save As**” to download the webpage to your PC.
 - Copy the content of the downloaded HTML file and paste it into *index.html* on **Server1** by clicking **Edit**, then save the changes.
 - Finally, import all downloaded objects (e.g., images, CSS files, etc.) into **Server1**, as shown in Figure 14. Note: You may need to update the file paths for these objects in the *index.html* file.

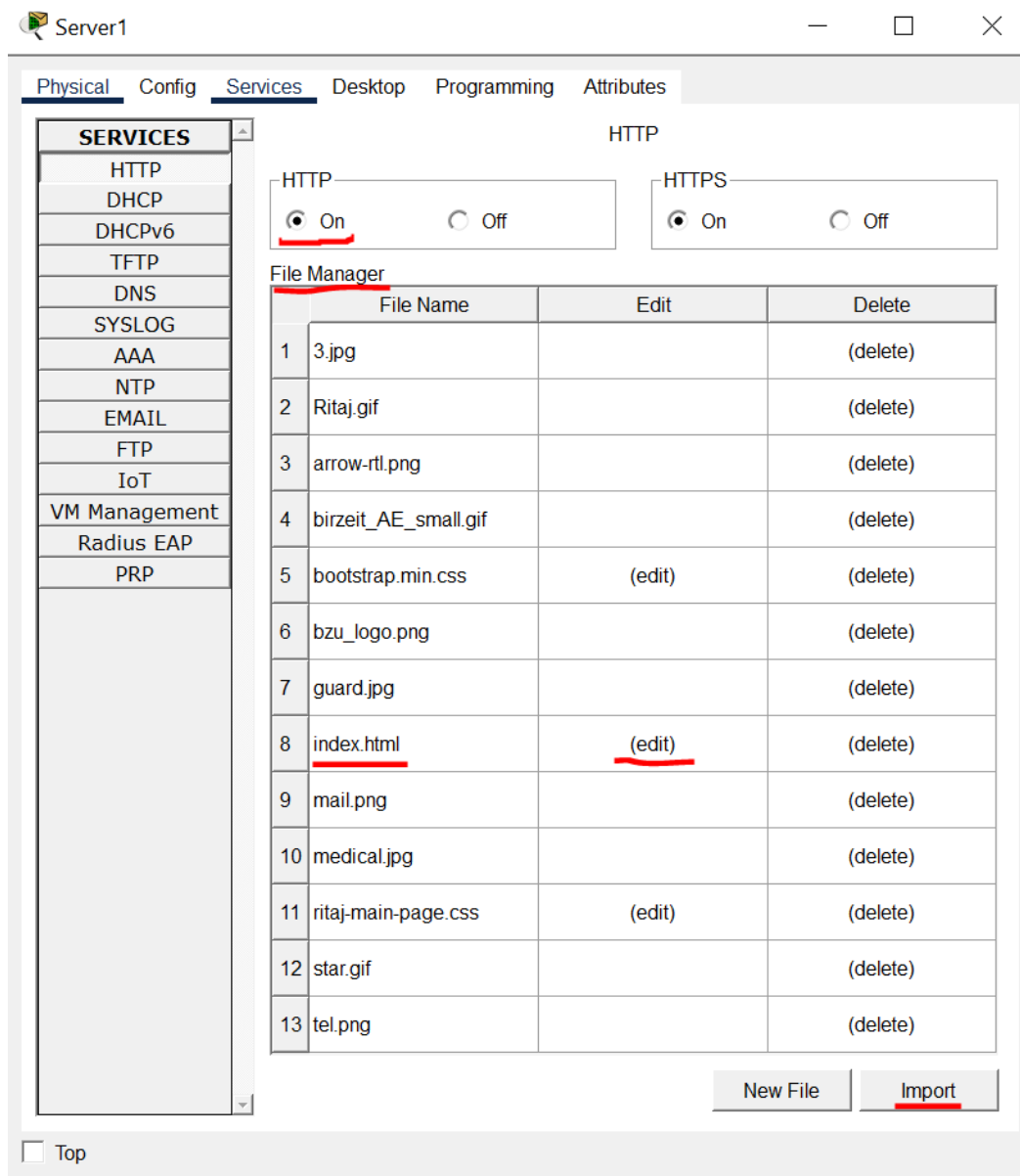


Figure 14: Configuring the Web server.

I) Configuring the DNS Server

Next, we will configure the DNS server (**Server2**) to resolve domain names to IP addresses, as shown in Figure 15.

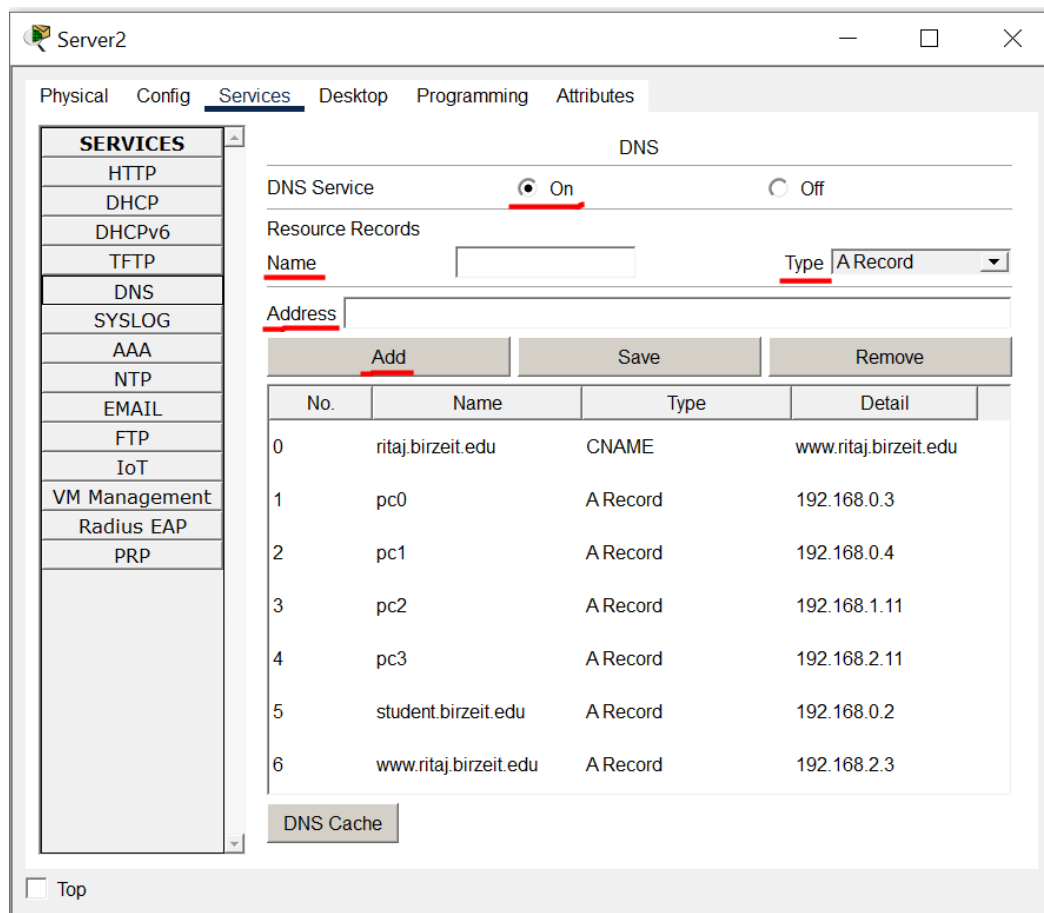


Figure 15: Configuring the DNS server.

- **Enable DNS on Server2:**

1. Open **Server2** and select the **Services** tab.
2. Under Services, select **DNS** and switch the DNS service to **On**.

- **Add DNS Entries:**

1. In the **Name** field, enter the domain name (e.g., www.ritaj.birzeit.edu).
2. In the **Address** field, enter the corresponding IP address.
3. Select the record **type** as “A Record”.
4. **Add** the entry to the DNS table.
5. Repeat these steps for any additional domain names (i.e., student.birzeit.edu) and PC devices.

Note: In real-world scenarios, PCs are not usually added to the DNS server. However, in this experiment, we add them for testing purposes so that packets can be sent using hostnames instead of IP addresses.

6. To allow the web server to also be reached using the URL “ritaj.birzeit.edu”, add a “CNAME” Record as shown in Figure 15.

J) Configuring the Email Server

In this section, we will configure the Email Server (**Server3**) to enable sending and receiving emails within the network, as shown in Figure 16.

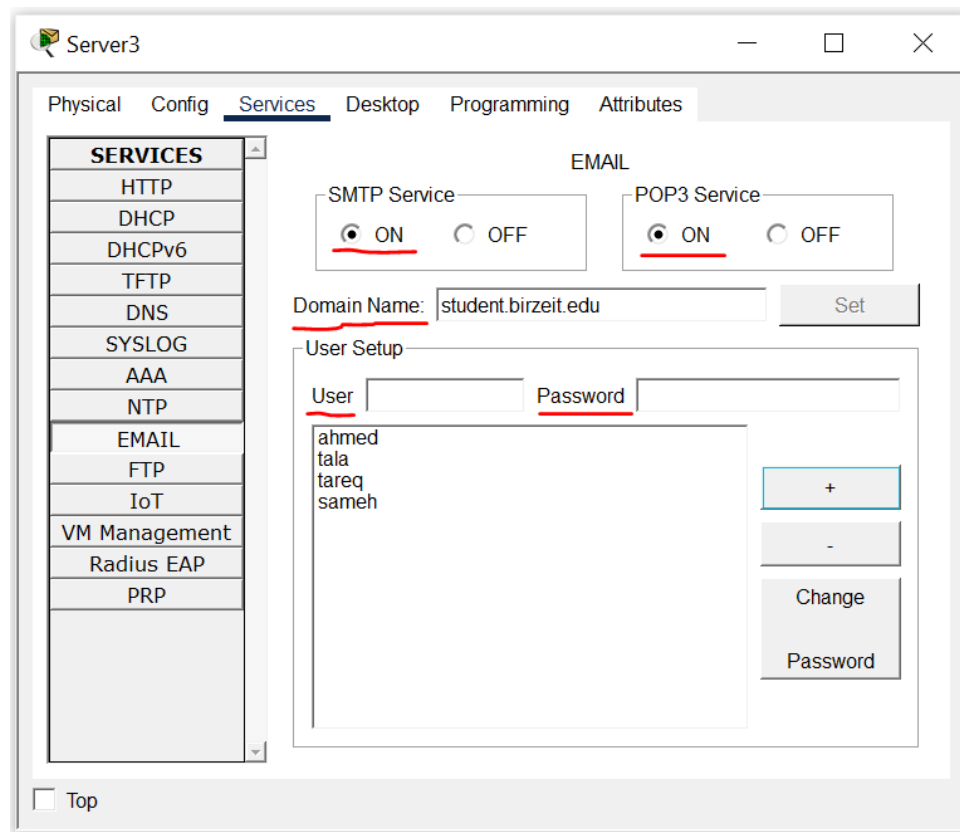


Figure 16: Configuring the Email server.

- **Enable Email Service on Server3:**

1. Go to **Server3** and select the **Services** tab.
2. Under Services, select **Email**, and turn **On** the **SMTP** and **POP3** services.

- **Add User Accounts:**

1. Set the **Domain Name** (mail server alias hostname) to “*student.birzeit.edu*”. Accordingly, each email address will follow the format: *User@DomainName* (e.g., ahmed@student.birzeit.edu).
2. Navigate to the **User Setup** section.
3. Create a user account by specifying the username and password. **Configure four user accounts (one for each PC)**. Students may choose the username and password for each account as they prefer.
4. Save the user entries.

- **Configure Email Clients on PCs:**

1. On each PC, go to **Desktop** → **Email**, then select **Configure Mail**.
2. Configure the email client on the PC using the following settings:
 - (a) **Your Name:** Full name
 - (b) **Email Address:** User@DomainName (e.g., *ahmed@student.birzeit.edu*)
 - (c) **Incoming Mail Server:** student.birzeit.edu
 - (d) **Outgoing Mail Server:** student.birzeit.edu
 - (e) **User Name:** User
 - (f) **Password:** Password

Figure 17 shows the configuration of the email client on **PC0** for the first account. Similarly, configure the email clients on **all other PCs**.

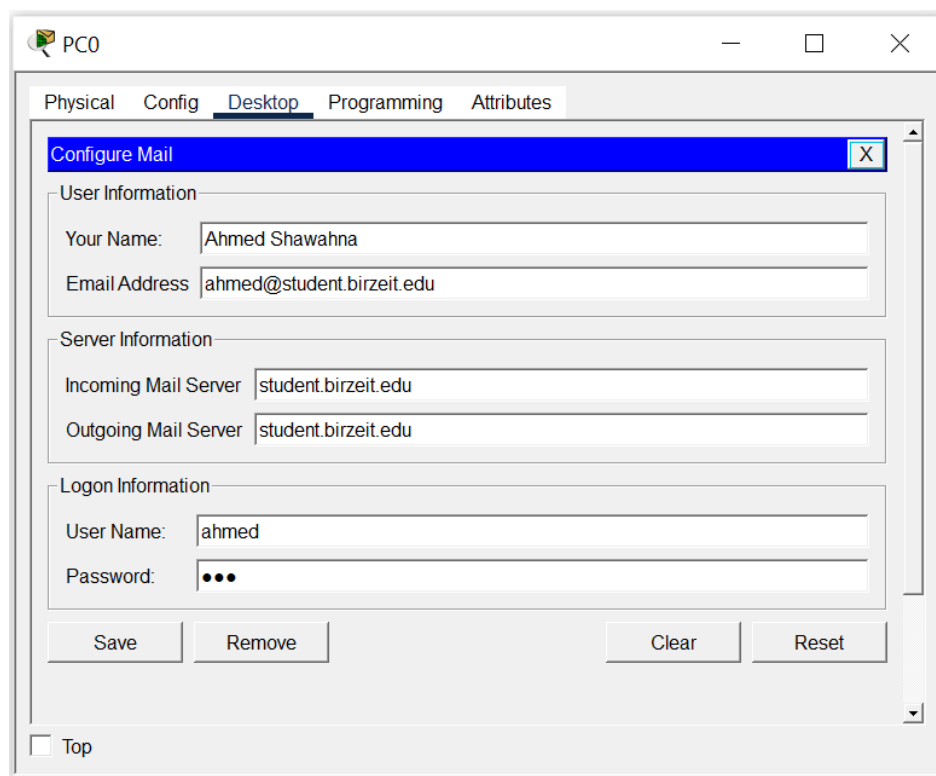


Figure 17: Configuring the Email client on PC0.

- **Sending an Email between Users:**

Now, we will send an email from user “**ahmed**” (on **PC0**) to user “**tala**” (on **PC1**).

1. Open the email client on **PC0**.
2. **Compose** a new email, set the recipient to “**tala@student.birzeit.edu**”, enter a subject and message (as shown in Figures 18 and 19), and click **Send**.
3. On **PC1**, open the email client. Verify that user “tala” has received the email by clicking **Receive** to display all received emails, as illustrated in Figure 20.

4. To reply, user “tala” can select the received email and click **Reply**.

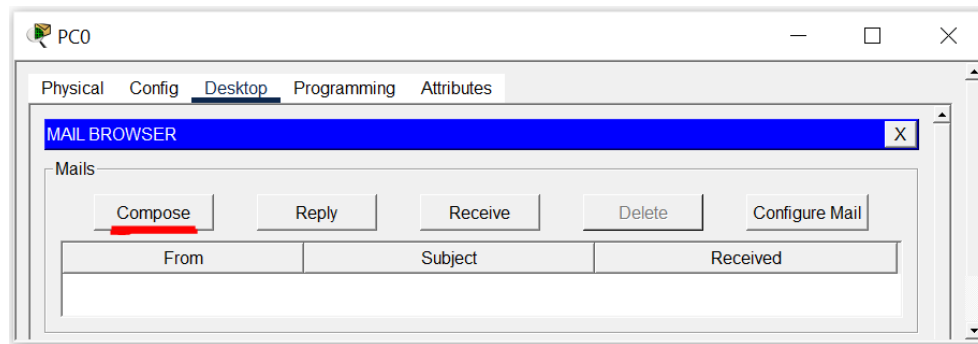


Figure 18: Compose a new email from user “ahmed” to user “tala”.

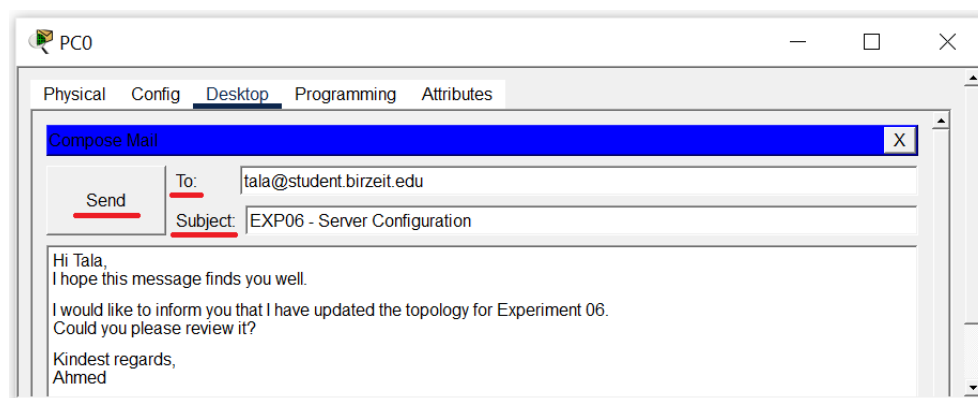


Figure 19: Sending an Email.

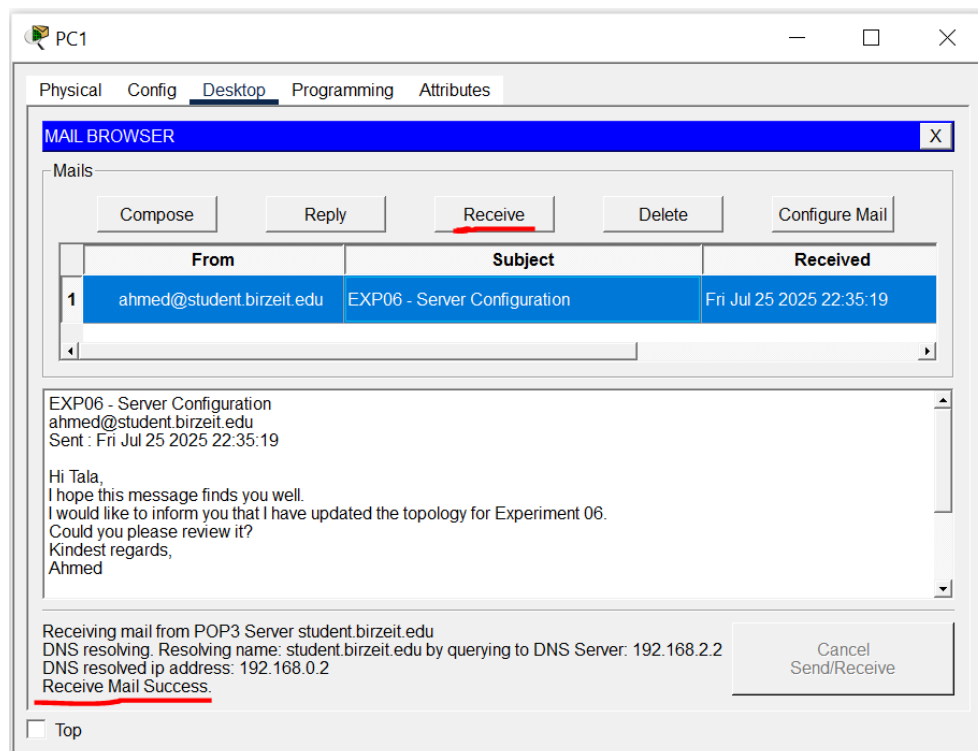
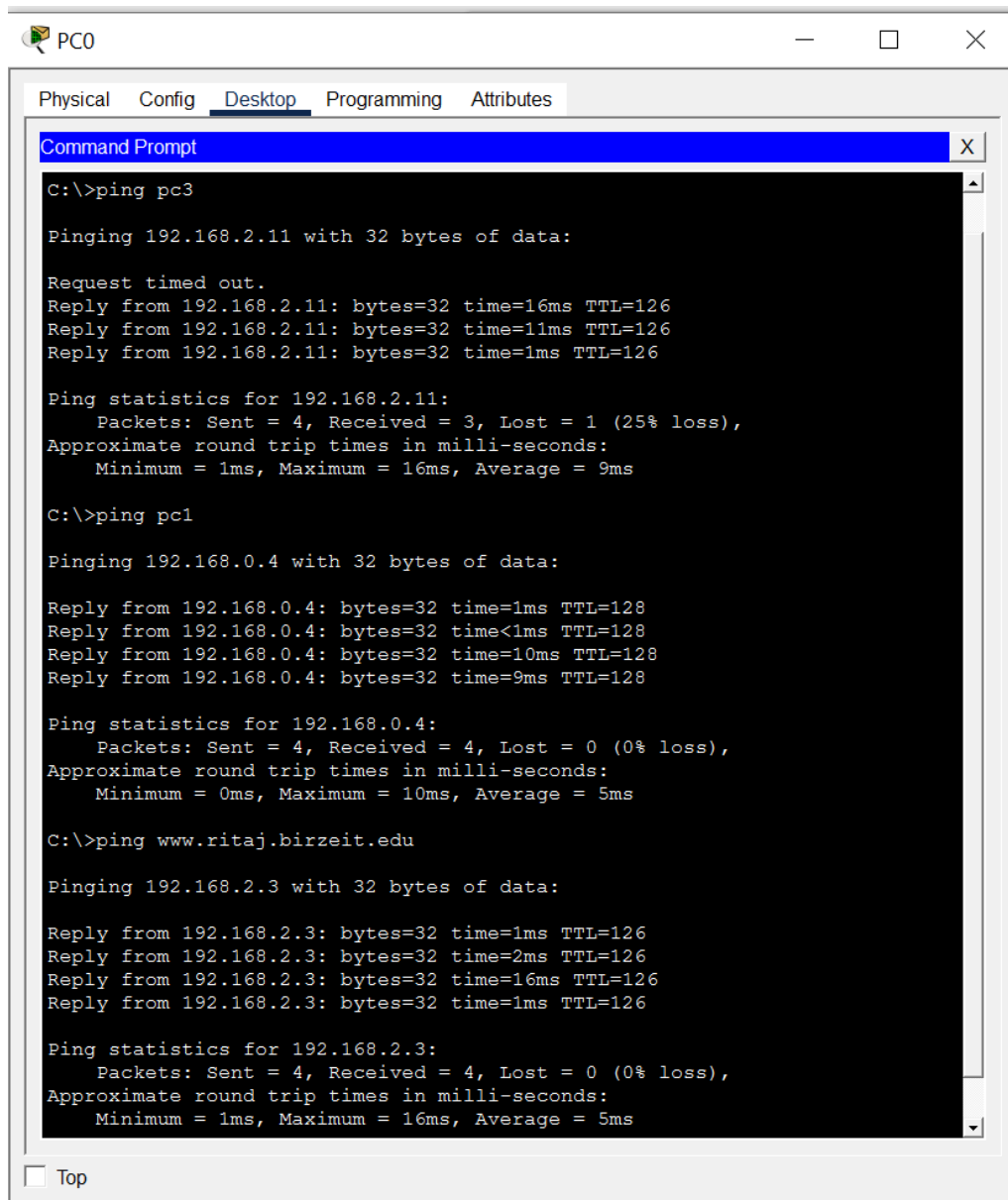


Figure 20: Show received Emails.

6.6.3 Testing and Verification

A) DNS Service Testing

- From any PC, open the **Command Prompt** and use the **ping** command to test connectivity to other PCs and servers using their hostnames, as shown in Figure 21.
- Verify that the domain names resolve to the correct IP addresses.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping pc3

Pinging 192.168.2.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.11: bytes=32 time=16ms TTL=126
Reply from 192.168.2.11: bytes=32 time=11ms TTL=126
Reply from 192.168.2.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 9ms

C:\>ping pc1

Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time=1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time=10ms TTL=128
Reply from 192.168.0.4: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping www.ritaj.birzeit.edu

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=2ms TTL=126
Reply from 192.168.2.3: bytes=32 time=16ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 5ms

Top
```

Figure 21: Pinging devices by their names.

B) Web and DNS Services Testing

- On any PC, open a **web browser**.
- Enter the IP address of **Server1** (**192.X.2.3**) or its domain names (**ritaj.birzeit.edu** and **www.ritaj.birzeit.edu**).

- Verify that the webpage hosted on **Server1** loads correctly, as shown in Figure 22.



Figure 22: Web server testing by domain name.

C) Email and DNS Services Testing

Using **Simulation Mode**, apply a filter for the **DNS**, **TCP**, **SMTP**, and **POP3** protocols. Then, send an email from the mail account on **PC2** to the mail account on **PC0**, and retrieve the email on **PC0**. Observe and follow the sequence of packets for:

- DNS query and reply.
- TCP three-way handshake (SYN, SYN+ACK, ACK).
- SMTP communication.
- POP3 communication.

Carefully document the sequence of observed packets, noting the order and type of each. Then, analyze and discuss your findings with your instructor or TA.

6.6.4 Packet Sniffing

We use the **Sniffer** device to understand how data flows through the network.

1. Enable Packet Capturing:

- Go to **Sniffer** → **GUI** → **Service**, and enable packet capturing.
- Select **Port0** as the interface that will be used to capture incoming packets.

2. Apply Filters:

The sniffer captures all protocols by default, to focus on specific network traffic:

- Click **Clear** to remove all previously captured packets.
- Click **Edit Filters** to select specific protocols, such as **DNS** and **ICMP**, for more precise analysis, as shown in Figure 23.

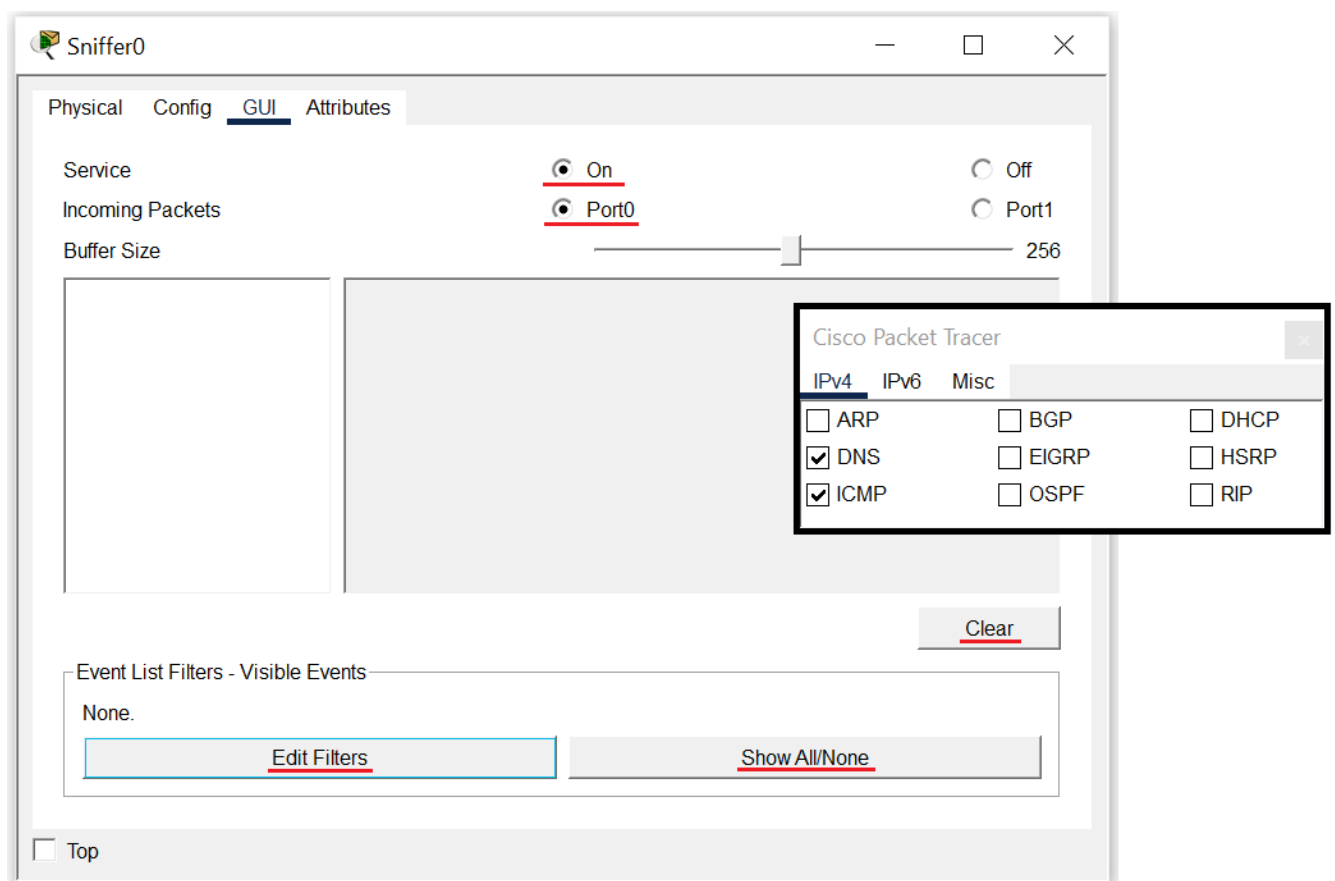


Figure 23: Applying filters to capture specific protocols.

3. Perform a Ping Using a Domain Name:

- From **PC1**, ping **PC3** using its domain name (e.g., **ping pc3**).
- The domain name is resolved by the DNS server, so the first packets sent are **DNS requests and replies** to obtain PC3's IP address.
- Once the IP address is resolved, **ICMP packets** are sent for the actual ping operation to test connectivity.

4. Analyze DNS and ICMP Packets:

- **DNS packets** display the name resolution process as PC1 queries the DNS server for PC3's IP address, as shown in Figure 24.
- **ICMP packets** verify connectivity between PC1 and PC3 after the domain name has been resolved, as shown in Figure 25.

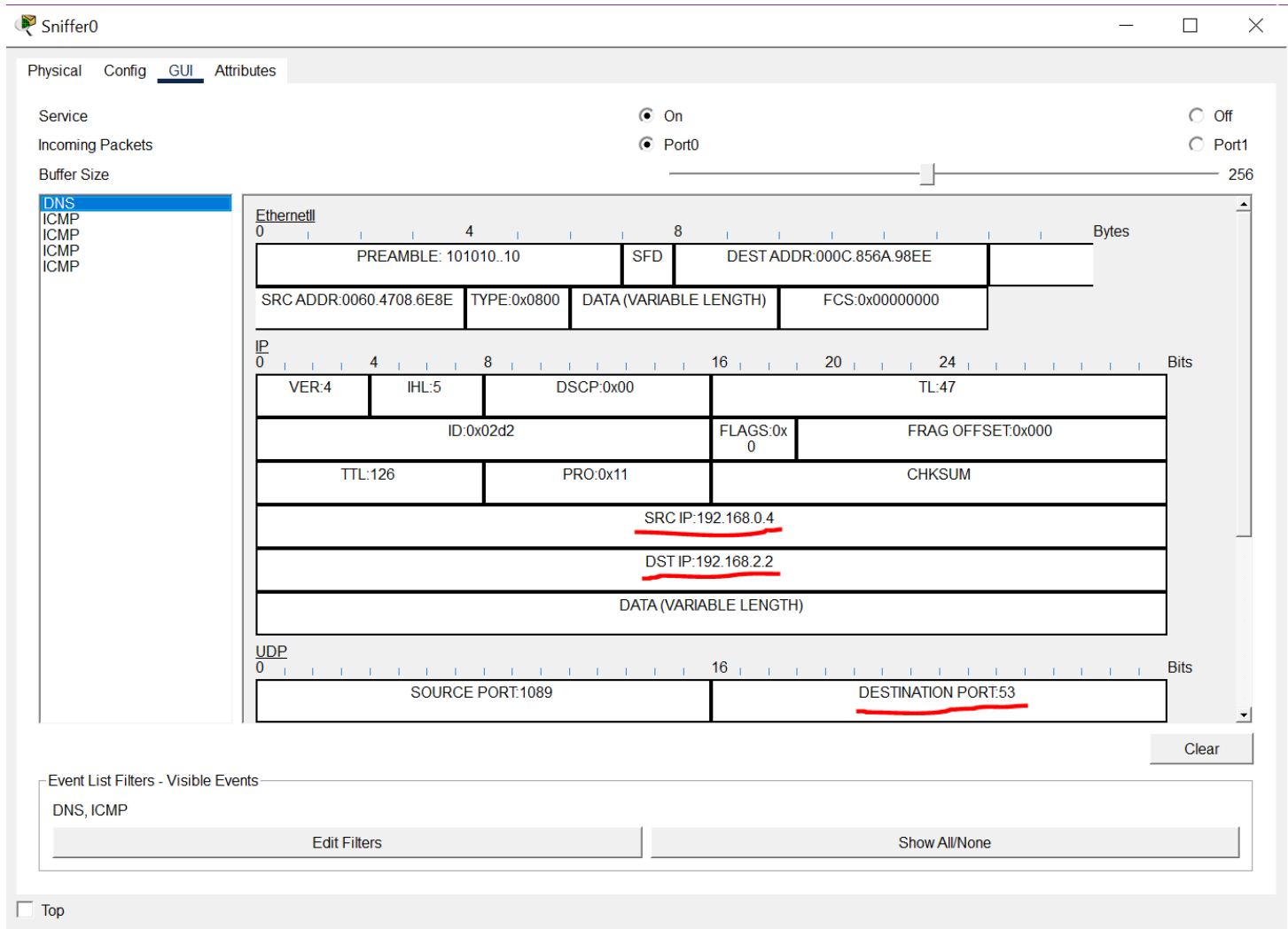


Figure 24: Packet details for DNS protocol.

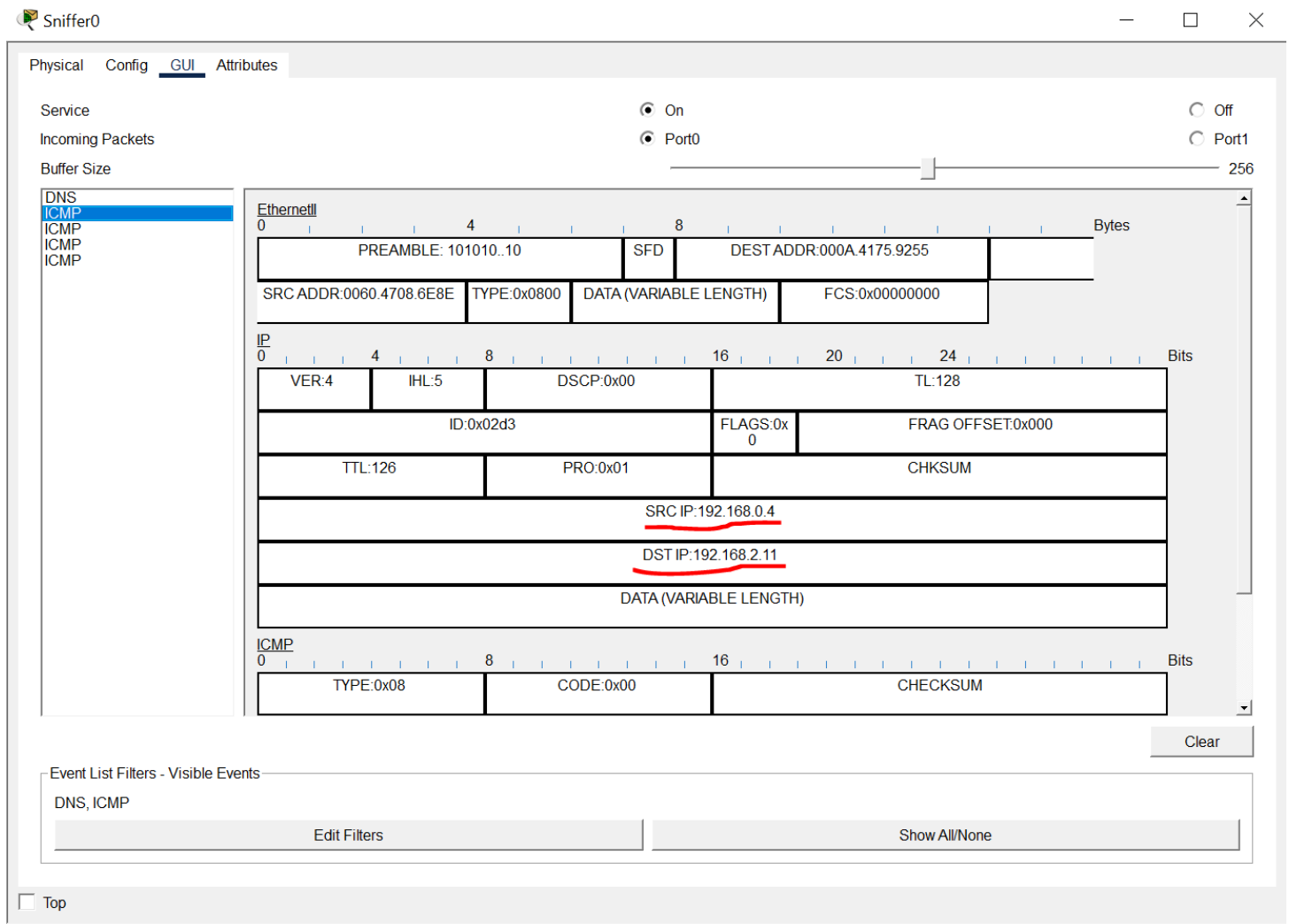


Figure 25: Packet details for ICMP protocol.

6.7 ToDo

This section will be provided by the instructor.